

Formalization of Gröbner Bases with Arbitrary Variable Types in Lean 4

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Abstract

We present a Lean 4 formalization of Gröbner basis theory over fields with arbitrary variable types, built on Mathlib’s polynomial infrastructure. Bases are represented as sets, and remainders are specified using finitely supported witnesses with degree bounds. This design supports finite and infinite bases uniformly, including polynomial division, Buchberger’s criterion, and the existence and uniqueness of reduced Gröbner bases. Monomial-order embeddings transport remainders and bases between variable types. A filter-based reconstruction theorem recovers reduced bases from restrictions to subrings under an eventual-coverage condition, extending reconstruction along countable ascending chains. Two separate developments demonstrate reuse of the library: tactics that certify polynomial and ideal computations performed by external computer algebra systems, and a formal solution over $\mathbb{C}$ to Iima–Yoshino Problem 2.3. The latter verifies an infinite reduced Gröbner basis and derives degreewise partition bijections establishing the partition form of the first Rogers–Ramanujan identity.

Keywords: Gröbner bases, Lean, Mathlib, formalization

1 Introduction

Gröbner bases provide a basic interface between commutative algebra and computation. They turn questions about polynomial ideals into questions about leading terms and reduction [4, 5]. Their algorithmic importance has motivated substantial developments in different proof assistants. There are formalizations of Gröbner basis theory that have been developed in the Isabelle/HOL proof assistant [18], including certified implementations of advanced methods such as Faugère’s F_4 algorithm [8] and signature-based Gröbner basis algorithm [13–15]. However, Gröbner basis theory has not yet been developed systematically in Lean 4 [17]. In Coq [2], early efforts focused on formalizing Buchberger’s algorithm and establishing the constructive existence of Gröbner bases [19, 24]. Additional developments include formalizations in ACL2 [16] and Mizar [21]. Despite this substantial body of work, Gröbner basis theory has not yet been developed systematically in Lean 4. Dermitzakis implemented S-polynomials and basic Gröbner basis constructions in Lean 3 [6]. While verifying the feasibility of such a development, this work remained largely isolated from Mathlib[23]. Poulsen and Guo explored an initial formalization of Gröbner basis theory, respectively, within Lean 3 [20] and the new Lean 4 ecosystem [12].

Many algebraic problems arise as compatible families $I_n \subseteq k[x_1, \dots, x_n]$ whose ambient rings grow with a parameter. Formalizing the type of variable index as a parameter instead of hard-coded $\mathbb{N}$ or $\{0, 1, \dots, n\}$ allows to unify the theory for different kinds of indices, including structured index types such as matrix positions. Especially finite-variable and infinite-variable polynomial ring and Gröbner basis can also be uniformly covered by our formalization by passing different arguments.

Infinite-variable Gröbner bases have applications in algebra, geometry, and combinatorics. Iima and Yoshino develop the theory for countably many variables, reconstruct reduced bases from finite-variable subrings, and use division to prove Schur-type partition identities [11]. Bruschek, Mourtada, and Schepers connect infinite Gröbner bases of arc ideals with the Rogers–Ramanujan identities [3]. In algebraic statistics, Hillar and Sullivant use monoidal Gröbner bases to prove stabilization results for Markov bases and the independent-set conjecture [10]; related equivariant-Noetherianity arguments give uniform finite descriptions of the Zariski closures of k -factor models and chirality varieties [7].

Our formalization treats classical and infinite-variable Gröbner bases within a common framework. Its main contributions are as follows.

- Building on Mathlib’s infrastructure, we formalize the Gröbner basis theory, including remainder, Gröbner basis, reduced Gröbner basis, and their important theorems, with arbitrary variable types and not necessarily finite Gröbner bases. Parts of this development have already been merged into Mathlib, and the formalization is designed to support long-term reuse at the library-level.
- We also provide theorems that can bridge the finite- and infinite-variable settings through monomial-order embeddings, including a generalized version of Iima–Yoshino’s construction of infinite reduced Gröbner basis [11, Theorem 1.12].
- We demonstrate how the library supports both certified computation and proofs involving infinite bases. Shen et al. [22] reuse its verification theorems in tactics for polynomial and ideal reasoning. Guo et al. [9] use its Gröbner-basis and remainder interfaces to formalize a solution over $\mathbb{C}$ to Iima–Yoshino Problem 2.3, including an infinite reduced basis and degreewise partition bijections.

Guo et al. [9] use our library in a separate formalization of a solution over $\mathbb{C}$ to Iima–Yoshino Problem 2.3. This application requires both infinitely many variables and an infinite Gröbner basis, and reuses various components of our formalization infrastructure.

Section 2 introduces the relevant Mathlib infrastructure, including some components developed in this work that have already been merged into Mathlib. Section 3 develops our formalization of Gröbner basis theory, covering the remainder relation, Buchberger’s criterion, and the existence and uniqueness of reduced Gröbner bases. Section 4 connects the finite and infinite variable settings through monomial-order embeddings and filter-based reconstruction. Section 5.2 presents two applications demonstrating the reuse of the library.

2 Formalization Infrastructure

Our Lean 4 formalization of Gröbner basis theory builds on Mathlib [23], including `Finsupp`, `MvPolynomial`, and `MonomialOrder`. This section introduces the underlying representations and explains some design choices that support our development.

2.1 Existing Mathlib Infrastructure

Let R be a commutative semiring and let σ be a type indexing the variables. We denote

$$P := R[X_i \mid i \in \sigma] \quad \text{and} \quad \mathcal{M}_\sigma := \sigma \rightarrow_0 \mathbb{N}.$$

They are written as `MvPolynomial` σ R and $\sigma \rightarrow_0 \mathbb{N}$ respectively in Lean. An element $a \in \mathcal{M}_\sigma$ is a finitely supported function from σ to $\mathbb{N}$. It is interpreted as an exponent vector and determines the monomial with exponent a

$$X^a := \prod_{i \in \sigma} X_i^{a(i)}.$$

The product is finite because

$$\text{supp}(a) := \{i \in \sigma \mid a(i) \neq 0\}$$

is finite.

For example, the monomial $X_1^3 X_4^5$ in $R[X_i \mid i \in \mathbb{N}]$ is represented by the exponent vector $e \in \mathbb{N} \rightarrow_0 \mathbb{N}$ given by

$$e(i) = \begin{cases} 3, & i = 1, \\ 5, & i = 4, \\ 0, & \text{otherwise.} \end{cases}$$

And a polynomial $p \in P$ is a finite sum of terms (scalar products of $\text{coeff}_p(a) \in R$ and X^a) with different exponents as follows, where $\text{coeff}_p : \mathcal{M}_\sigma \rightarrow_0 R$ is also a finitely supported function

$$p = \sum_{a \in \mathcal{M}_\sigma} \text{coeff}_p(a) X^a,$$

where each non-zero $\text{coeff}_p(a) X^a$ is a term of p , and $\text{supp}(p) := \{a \in \mathcal{M}_\sigma \mid \text{coeff}_p(a) \neq 0\}$ is called the support of p .

This representation imposes no finiteness (Fintype in Lean) assumption on σ , so the same type uniformly represents either finite-variable polynomial rings such as $R[X_0, X_1, X_2]$ or infinite-variable polynomial rings such as $R[X_0, X_1, \dots]$.

The exponent type $\mathcal{M}_\sigma$ also carries the pointwise partial order

$$a \leq b \iff \forall i : \sigma, a(i) \leq b(i).$$

This order corresponds to monomial divisibility: $a \leq b$ precisely when X^a divides X^b , in which case

$$X^b = X^{b-a} X^a.$$

This pointwise order plays a different role from the monomial order introduced below: the former captures divisibility, whereas the latter is used to compare monomials and select leading terms.

2.2 Monomial Orders and Leading Terms

On $\mathcal{M}_\sigma$, we distinguish between two different orders. The notation $a \leq b$ denotes the componentwise order introduced in Section 2.1, whereas $a \preceq_m b$ denotes comparison with respect to a chosen monomial order m .

Definition 1 (Monomial order). A monomial order on $\mathcal{M}_\sigma$ is a relation $\preceq_m$ satisfying the following conditions:

1. $\preceq_m$ is a linear order whose strict part $\prec_m$ is well founded;
2. for all $a, b, c \in \mathcal{M}_\sigma$,

$$a \preceq_m b \implies a + c \preceq_m b + c;$$

3. 0 is the least element:

$$0 \preceq_m a \quad \text{for every } a \in \mathcal{M}_\sigma.$$

In Mathlib, this notion is formalized with the structure `MonomialOrder`. Different monomial orders may be used simultaneously on the same exponent type, so Mathlib does not equip the monomial order as the typeclass order on $\mathcal{M}_\sigma$. Instead, it associates to m an ordered additive type $\mathcal{M}_\sigma^{(m)}$ and an additive equivalence

$$\iota_m : \mathcal{M}_\sigma \simeq_+ \mathcal{M}_\sigma^{(m)}.$$

The selected monomial order is the pullback of the order on $\mathcal{M}_\sigma^{(m)}$:

$$a \preceq_m b \iff \iota_m(a) \leq \iota_m(b). \quad (1)$$

Definition 2 (Lexicographic order). Suppose that σ is equipped with a linear order where $>$ is well-founded. For distinct $a, b \in \mathcal{M}_\sigma$, define the set of indices at which they differ by

$$D(a, b) := \{i \in \text{supp}(a) \cup \text{supp}(b) \mid a(i) \neq b(i)\}.$$

This is a finite nonempty set. Let i_0 be its least element. The lexicographic order is defined by

$$a \prec_{\text{lex}} b \iff a(i_0) < b(i_0).$$

Finite support guaranties that the first differing index i_0 exists even when σ is infinite. To obtain a monomial order, one additionally requires the relation on σ to be well founded. Mathlib packages the resulting construction as Lexicographic order.

Fix a monomial order m , and let

$$p = \sum_{a \in \text{supp}(p)} \text{coeff}_p(a) X^a$$

be a nonzero polynomial. Its degree with respect to m is

$$\deg_m(p) := \max_{\preceq_m} \text{supp}(p),$$

which is well defined because $\text{supp}(p)$ is finite and nonempty. Its leading coefficient and leading monomial are defined by

$$\text{LC}_m(p) := \text{coeff}_p(\deg_m(p)), \quad \text{LM}_m(p) := X^{\deg_m(p)},$$

respectively, and its leading term is

$$\text{LT}_m(p) := \text{LC}_m(p) \text{LM}_m(p). \quad (2)$$

We omit the subscript m when the monomial order is clear from the context.

2.3 Degree with Bottom Element

In the previous formalization in Mathlib, $\deg$ is formalized as a function $P \rightarrow \mathcal{M}_\sigma$, in which the degree of zero polynomial is defined 0.

```
variable {m : MonomialOrder σ} in
def degree (f : MvPolynomial σ R) : σ →0 ℕ
```

It is convenient and reasonable, because $\mathcal{M}_\sigma$ is the type of exponent vector and is widely used in related APIs, and 0 is the least element in $\mathcal{M}_\sigma$ with respect to both pointwise partial order and any monomial order, considering that we will hope $\deg(0)$ not greater than $\deg(1) = 0$. However, this convention fails to satisfy some desirable properties. In particular, 0 and constant polynomial cannot be distinguished by their degrees, and as a result the additive property when R is a domain that

$$\deg(fg) = \deg(f) + \deg(g)$$

does not hold without additional assumptions. When $f = 0$ and g isn't a constant polynomial, then $\deg(fg) = 0$ while $\deg(f) + \deg(g) = \deg(g) \neq 0$.

To resolve this issue in Lean, we provide a variant of degree function where the codomain is extended to $\mathcal{M}_\sigma \cup \{-\infty\}$ (WithBot $(\sigma \rightarrow_0 \mathbb{N})$ in Lean), with a bottom element $-\infty$ ($\perp$ in Lean) strictly smaller than every exponent vector. This allows us to separate the degree of the zero polynomial from that of constant polynomials:

- $\deg^\perp(0) = -\infty$;
- $\deg^\perp(g) = \deg(g)$, when $g \neq 0$, and specially $\deg^\perp(c) = 0$ if c is a non-zero constant polynomial.

Under this convention, $\deg^\perp(0) < \deg^\perp(c)$ is valid and the classical additive properties of degree can be recovered, since $-\infty + a = -\infty$ for all a .

We formalize the degree function with a bottom element as follows.

```
variable (m : MonomialOrder σ) in
def withBotDegree (f : MvPolynomial σ R) : WithBot (σ →0 ℕ) :=
  f.support.image m.toSyn |>.max.map m.toSyn.symm
```

When we need to distinguish between zero and non-zero constant polynomials with their degrees, we tend to use $\deg^\perp$ instead of stating by cases with $\deg$. While in other cases, we still tend to use $\deg$ (degree) in Lean, since most APIs involved degree are stated in degree.

3 Formalization of Gröbner Bases Theory

In this section, we formalize the connection between polynomial reduction and Gröbner bases. We first introduce a remainder relation that specifies the properties of division independently of a particular division procedure. We then define Gröbner bases through leading-term ideals and establish their characterization in terms of remainders. This characterization provides the basis for our proof of Buchberger's criterion. Throughout, neither the variable type nor the set of divisors is assumed to be finite.

3.1 Remainders

Definition 3 (Remainder). Let f be a multivariate polynomial and B a set of multivariate polynomials over $P := R[X_i | i \in \sigma]$ where R is a commutative ring, and fix a monomial order m . Suppose that there exists a family of polynomial $\{g_b\}_{b \in B}$ with only finite non-zero polynomials, s.t.

- (1) $f = \sum_{b \in B} g_b b + r$,
- (2) $\deg^\perp(g_b) + \deg^\perp(b) \preceq_m \deg^\perp(f)$ for all $b \in B$,
- (3) $\deg(b) \not\leq c$ for all $c \in \text{supp}(r)$ and $b \in B \setminus \{0\}$.

r is called a remainder of f on division by B , and $\{g_b\}_{b \in B}$ is called a quotient family. We also write $f \xrightarrow{B} r$ for the remainder on division of f by polynomials in set B .

We formalize it in Lean as follows, where the quotient family is formalized with $B \rightarrow_0 \text{MvPolynomial } \sigma R$.

```
def IsRemainder :=
  (∃ (g : B →0 MvPolynomial σ R),
    -- (1)
    f = Finsupp.linearCombination _ (fun (b : B) ↦ b.val) g + r
    ∧
    -- (2)
    ∀ (b : B), m.toWithBotSyn (m.withBotDegree b.val) +
      m.toWithBotSyn (m.withBotDegree (g b)) ≤
      m.toWithBotSyn (m.withBotDegree f)
    ∧
    -- (3)
    ∀ c ∈ r.support, ∀ b ∈ B, b ≠ 0 → ¬ (m.degree b ≤ c))
```

This concept is from (while weaker than) reducing terms of f and the result of last reduction by leading terms of polynomials $b \in B$, until all terms are no longer reducible. Formally, let $f_0 = f$, and $f_{i+1} = f_i - c_i x^{\alpha(i)} b_i$ (resulting in (1)) if there exist $b_i \in B$ and monomial $c_i x^{\alpha(i)}$, s.t. $c_i x^{\alpha(i)}$ LT(b_i) is a term of f_i (resulting in (2)). From the well-foundedness of the monomial order, there must be a f_n obtained from reduction s.t. there exists no f_{n+1} , which results in (3) and gives

a remainder f_n of f if R is a field or $\text{LC}(B)$ are all invertible. Generally a given f may be reduced by B to different remainders with different choices of b_i .

The following variants of (2) in the definition were considered and tried in our previous formalization:

- (2.a) $\deg(g_b b) \preceq \deg(f)$ before we formalized $\deg^\perp$,
- (2.b) $\deg^\perp(g_b b) \preceq \deg^\perp(f)$.

When we started our formalization, we considered only the cases where R is a field or at least a domain, where (2.a) is sufficient. When R is a domain, the only difference between (2) and (2.a) is the edge case that $f = 0$ and two distinct $b_1, b_2 \in B$ are non-zero constant polynomials. In this case, (2.a) allows g_1 and g_2 to be non-zero constants kb_2 and $-kb_1$ where $k \in R \setminus \{0\}$, which does not match the idea of reducing f by $b_1, b_2 \in B$. The definition with (2.a) is still equivalent to the one with (2), though. This is because no matter in the (2) version or (2.a) version, the remainder of 0 must be 0 with the limit of (1) and (3).

We hope that the formalization can be more general in the hypothesis of R with most properties still preserved. However, when it comes to the cases where R is not a domain, the difference between (2) and (2.a) matters. Two versions of remainder definition are no longer equivalent. To let the degree of 0 be strictly less than degree of non-zero constant, avoiding the cases where 0 seems reduced by non-zero polynomial, we also formalized $\deg^\perp$ tried (2.b). While the common problem of (2.a) and (2.b) is that, when the leading coefficients of some polynomials g_b are zero divisors, their leading terms multiplied by non-zero monomials may give 0, and the leading term thus “disappears” (2.a) and (2.b). In this case f is reduced by another term of b instead of its leading term. For example, let $R = \mathbb{Z}/4\mathbb{Z}$, $f = 2x$, and $B = \{2x + 1\}$, and then $0 = f - 2x(2x + 1)$ while $\deg^\perp(2xg) = \deg^\perp(f)$. To exclude these cases, we had to add hypothesis to some theorems that $\text{LC}(B)$ are all non-zero-divisors.

Eventually, we choose (2), which compares the degree of f with the degree of g_b and b instead of the degree of their product, so that the leading term will not be removed by the coefficient. The non-zero-divisor hypothesis of many theorems thus gets removed. Is it worth mentioning that the difference between $\deg^\perp$ and $\deg$ plays a role here, to differentiate between the degrees of non-zero constants and zero, and let $\deg^\perp(0) + \deg^\perp(b) = -\infty$. These require stated case-by-case if $\deg$ is used.

3.2 Gröbner Bases

Fix a monomial order m on $\mathcal{M}_\sigma$, let R be a commutative ring, and let $I \subseteq R[X_i | i \in \sigma]$ be an ideal. We define Gröbner basis G of I through equality between the leading term ideals of G and I and then relate this definition to the remainder relation.

Definition 4 (Gröbner Basis). A subset $G \subseteq I$ is said to be a Gröbner basis of I if the leading terms of the elements of

G generate the leading term ideal of I , i.e.

$$\langle \text{LT}(G) \rangle = \langle \text{LT}(I) \rangle,$$

where $\text{LT}(G)$ and $\text{LT}(I)$ denote the set of leading terms of elements of G and I , respectively. Notice that G is not necessarily finite in our definition.

The formal definition is as follows:

```
def IsGroebnerBasis {R : Type*} [CommSemiring R] (G : Set (
  MvPolynomial σ R))
  (I : Ideal (MvPolynomial σ R)) :=
  G ⊆ I ∧ Ideal.span (m.leadingTerm '' ↑I) = Ideal.span (m.
    leadingTerm '' G)
```

Having established the definition of Gröbner bases, we turn to the question of their existence. While ideal I is always a trivial infinite Gröbner basis, a finite Gröbner basis is important for application of Gröbner basis with concrete polynomials. The existence of a finite Gröbner basis of an ideal is can be showed if the ring $R[X_i | i \in \sigma]$ is Noetherian i.e. R is Noetherian and σ is finite, by choosing a $p \in I$ for each polynomial q in a generating set of $\text{LT}(I)$ s.t. $\text{LT}(p) = \text{LT}(q)$.

Beyond existence results, Gröbner bases play a central role in solving the ideal membership problem. A key result is presented in theorem 5.

Theorem 5. If $G \subseteq I$ is a Gröbner basis of I , then for any $p \in R[X_i | i \in \sigma]$, a remainder of p upon division by G is zero if and only if $p \in I$.

```
theorem isRemainder_zero_iff_mem_ideal {p : MvPolynomial σ R}
  {G : Set (MvPolynomial σ R)} {I : Ideal (MvPolynomial σ R)}
  (h : m.IsGroebnerBasis G I) : m.IsRemainder p G 0 ↔ p ∈ I
```

It follows that $p \in I$ can be expressed as a $R[X_i | i \in \sigma]$ -linear combination of elements of G , so $G \subseteq I \subseteq \langle G \rangle$, i.e. $\langle G \rangle = I$.

The following proposition characterizes the Gröbner bases in terms of remainders and provides a criterion for deciding ideal membership.

Theorem 6. Let I be an ideal of $R[X_i | i \in \sigma]$. A set G is a Gröbner basis of I if and only if $G \subseteq I$ and, for every $f \in I$, a remainder of f upon division by G is 0.

```
theorem iff_subset_ideal_and_isRemainder_zero
  (G : Set (MvPolynomial σ R)) (I : Ideal (MvPolynomial σ R)) :
  m.IsGroebnerBasis G I ↔ G ⊆ I ∧ ∀ p ∈ I, m.IsRemainder p G 0
```

Finally, an important property of Gröbner bases is that they provide a unique remainder for elements modulo the ideal they generate. Unlike a general multivariate division, where the remainder typically depends on the order of the divisors, division by a Gröbner basis guarantees a unique remainder.

Theorem 7. Let G be a Gröbner basis of I , and let $\text{LC}(G)$ be all invertible. Then for any polynomial p , the remainder of p upon division by G is unique.

```

theorem existsUnique_isRemainder {G : Set (MvPolynomial σ R)}
  {I : Ideal (MvPolynomial σ R)}
  (h : m.IsGroebnerBasis G I)
  (hG : ∀ g ∈ G, IsUnit (m.leadingCoeff g)) (p : MvPolynomial σ
    R) :
  ∃! (r : MvPolynomial σ R), m.IsRemainder p G r

```

3.3 Buchberger's Criterion

To formally verify Buchberger's criterion in Lean 4, we first introduce S-polynomials. Our definition of the S-polynomial differs slightly from the standard formulations in [1, 5]. We adopt this variant to obtain a more natural implementation in Lean, in particular avoiding the need to assume invertibility of leading coefficients when stating a S-polynomial.

Definition 8 (S-polynomial). Let $f, g \in k[X_i | i \in \sigma]$. Define the S-polynomial by

$$S(f, g) = \frac{LC(g)x^\gamma}{LM(f)} \cdot f - \frac{LC(f)x^\gamma}{LM(g)} \cdot g,$$

where x^γ denotes the least common multiple of $LM(f)$ and $LM(g)$.

In particular, $S(0, g) = S(f, 0) = 0$.

Since when $f, g \neq 0$ this formulation differs from the standard definition $\frac{x^\gamma}{LT(f)}f - \frac{x^\gamma}{LT(g)}g$ only by multiplication by a nonzero scalar factor $LC(f)LC(g)$, it does not affect the validity of Buchberger's criterion with the standard one.

We now give the formal definition of the S-polynomial following definition 8.

```

noncomputable def sPolynomial (f g : MvPolynomial σ R) :
  MvPolynomial σ R :=
  monomial (m.degree g - m.degree f) (m.leadingCoeff g) * f -
  monomial (m.degree f - m.degree g) (m.leadingCoeff f) * g

```

Before proving Buchberger's criterion, we first establish a lemma characterizing the cancellation of leading terms. If $f \neq 0$ and $\deg^\perp(f) = \deg^\perp(g)$, then $\deg^\perp(S(f, g)) < \deg^\perp(f)$. More generally, if a k -linear combination of polynomials with the same degree exhibits cancellation of leading terms, then it can be expressed as a k -linear combination of their S-polynomials.

```

lemma sPolynomial_decomposition {d : m.syn} {ι : Type*}
  {B : Finset ι} {g : ι → MvPolynomial σ R}
  (hd : ∀ b ∈ B,
    (m.toSyn <| m.degree <| g b) = d ∧ IsUnit (m.leadingCoeff <|
      g b) ∨ g b = 0)
  (hfd : (m.toSyn <| m.degree <| ∑ b ∈ B, g b) < d) :
  ∃ (c : ι → ι → R),
    ∑ b ∈ B, g b = ∑ b₁ ∈ B, ∑ b₂ ∈ B, (c b₁ b₂) · m.sPolynomial
      (g b₁) (g b₂)

```

Then we attempt to formalize Buchberger's criterion, which characterizes Gröbner bases via the reduction of S-polynomials of critical pairs.

Theorem 9 (Buchberger's Criterion). Assume all polynomials in a set $G \subseteq R[X_i | i \in \sigma]$ have invertible leading coefficients. Then G is a Gröbner basis of an ideal I if and only if $G \subseteq I$ and for all $f, g \in G$, 0 is a remainder of S-polynomial $S(f, g)$ on division of G .

Here's an example of using Buchberger's criterion.

Example 1. Consider the Cyclic3 ideal $I \subset \mathbb{Q}[x_0, x_1, x_2]$ with lexicographic order $x_0 \succ x_1 \succ x_2$, generated by

$$\{f_1 = x_0 + x_1 + x_2, \quad f_2 = x_0x_1 + x_1x_2 + x_2x_0, \quad f_3 = x_0x_1x_2 - 1\}.$$

A Gröbner basis for I is

$$G = \{g_1 = x_0 + x_1 + x_2, \quad g_2 = x_1^2 + x_1x_2 + x_2^2, \quad g_3 = x_2^3 - 1\}.$$

By Buchberger's criterion, it suffices to check that for all pairs $g_i, g_j \in G$, the S-polynomial $S(g_i, g_j)$ reduces to 0 modulo G . As an illustration, consider (g_1, g_2) . We have $LM(g_1) = x_0$, $LM(g_2) = x_1^2$, and $\text{lcm}(LM(g_1), LM(g_2)) = x_0x_1^2$, hence

$$\begin{aligned}
 S(g_1, g_2) &= \frac{x_0x_1^2}{x_0} g_1 - \frac{x_0x_1^2}{x_1^2} g_2 \\
 &= x_1^2(x_0 + x_1 + x_2) - x_0(x_1^2 + x_1x_2 + x_2^2) \\
 &= -x_0x_1x_2 - x_0x_2^2 + x_1^3 + x_1^2x_2 \\
 &= (-x_1x_2 - x_2^2)g_1 + (x_1 + x_2)g_2.
 \end{aligned}$$

Therefore, $S(g_1, g_2)$ reduces to 0 modulo G . The remaining pairs are checked similarly.

We now formalize Buchberger's criterion, as stated in theorem 9.

```

theorem iff_ideal_eq_span_and_isGroebnerBasis_span {R} [CommRing R]
  {I : Ideal (MvPolynomial σ R)} {G : Set (MvPolynomial σ R)}
  (hG : ∀ g ∈ G, IsUnit (m.leadingCoeff g)) :
  m.IsGroebnerBasis G I ↔ I = Ideal.span G ∧
  ∀ (g₁ g₂ : G), m.IsRemainder (m.sPolynomial g₁ g₂ :
    MvPolynomial σ R) G 0

```

3.4 Reduced Gröbner Bases

Since the definition of Gröbner basis only requires that the leading terms of its elements generate the leading term ideal $\langle LT(I) \rangle$, redundant generators may occur. For example, if $g_1, g_2 \in G$, $LM(g_1)$ divides $LM(g_2)$, then g_2 does not contribute any new information. Reduced Gröbner basis is a special kind of Gröbner basis that can serve as a unique representative in Gröbner bases of each ideal $I \subseteq k[X_i | i \in \sigma]$.

Definition 10 (Reduced Gröbner Basis). Fix a monomial order on $\mathcal{M}$. A subset $G \subseteq I$ is called a reduced Gröbner basis of the ideal I if the following conditions hold:

- (1) G is a Gröbner basis of I .
- (2) Every $p \in G$ is monic.
- (3) For all distinct $p, q \in G$, no monomial appearing in p is divisible by $LM(q)$.

We now formalize the notion of a reduced Gröbner basis in Lean, following definition 10:

```

def IsGroebnerBasis.IsReduced (hG : m.IsGroebnerBasis G I) :=
  (∀ p ∈ G, m.Monic p) ∧ ∀ p ∈ G, m.IsRemainder p (G \ {p}) p

```

Example 1 (continued). Consider $G' = \{g'_1, g'_2, g'_3\}$, where

$$g'_1 = x_0 + x_1^2 + x_1x_2 + x_1 + x_2^2 + x_2.$$

Then G' is a Gröbner basis of I , but it is not reduced: g'_1 contains the monomial x_1^2 , which is divisible by $\text{LM}(g_2) = x_1^2$. Replacing g'_1 with $g_1 = g'_1 - g_2$ removes this reducible term and yields the reduced Gröbner basis $G = \{g_1, g_2, g_3\}$ (see section 3.3).

The stronger condition in definition 10 yields the following fundamental property of reduced Gröbner bases.

Theorem 11. *Every ideal $I \subseteq k[X_i | i \in \sigma]$ admits a unique reduced Gröbner basis.*

```
theorem IsReduced.uniqueExists {k} [Field k] (I : Ideal (
  MvPolynomial σ k)) :
  ∃! (B : Set (MvPolynomial σ k)), ∃ (h : m.IsGroebnerBasis B I),
  h.IsReduced
```

The following version is stated for general commutative rings rather than fields, under the additional hypothesis that the leading coefficients of the Gröbner basis are units.

```
theorem IsReduced.uniqueExists_of_isGroebnerBasis {R} [Nontrivial R
  ] [CommRing R]
  {G : Set (MvPolynomial σ R)} {I : Ideal (MvPolynomial σ R)}
  (hG : m.IsGroebnerBasis G I) (hG' : ∀ g ∈ G, IsUnit (m.
    leadingCoeff g)) :
  ∃! (B : Set (MvPolynomial σ R)), ∃ (h : m.IsGroebnerBasis B I),
  h.IsReduced
```

4 Infinite-Variable Gröbner Bases

In this section, we explain how Gröbner bases in polynomial rings with infinitely many variables can be characterized via finite-variable reductions. Let σ and σ' be variable index types, and write two polynomial rings

$$S = k[X_i | i \in \sigma], \quad S' = k[x_i]_{i \in \sigma'}.$$

Let $\mathcal{M} := (\sigma \rightarrow_0 \mathbb{N})$ and $\mathcal{M}' := (\sigma' \rightarrow_0 \mathbb{N})$ be the corresponding types of exponent vectors, and denote monomial orders on $\mathcal{M}$ and $\mathcal{M}'$ as $(\mathcal{M}, \leq_{\mathcal{M}})$ and $(\mathcal{M}', \leq_{\mathcal{M}'})$, respectively.

A key prerequisite for relating these two settings is the compatibility of monomial orders. We now introduce the notion of an embedding of monomial orders.

Definition 12 (Monomial Order Embedding). Given two monomial orders $(\mathcal{M}, \leq_{\mathcal{M}})$ and $(\mathcal{M}', \leq_{\mathcal{M}'})$, a function $f : \sigma' \rightarrow \sigma$ is an order embedding if f satisfies

- (1) The function f is injective.
- (2) Denote the lift from $\mathcal{M}'$ to $\mathcal{M}$ via f as $f' : \mathcal{M}' \rightarrow \mathcal{M}$. For all u, v in $\mathcal{M}'$, if $u \leq_{\mathcal{M}'} v$, then $f'(u) \leq_{\mathcal{M}} f'(v)$.

We formalize definition 12 in Lean as follows:

```
structure Embedding extends σ' ↪ σ where
  monotone' : Monotone (m.toSyn ∘ Finsupp.mapDomain toFun ∘ m'.
    toSyn.symm)
```

Given a monomial order on $\mathcal{M}$, any injective map $f : \sigma' \rightarrow \sigma$ induces a monomial order on $\mathcal{M}'$. This order is obtained by pulling back the order on $\mathcal{M}$ along the embedding of exponent vectors induced by f

```
def ofInjective.Syn (m : MonomialOrder σ) {σ' : Type*} (f : σ' →
  σ) := σ' →_0 ℕ
noncomputable def ofInjective {σ' : Type*} {f : σ' → σ} (hf : f.
  Injective) :
  MonomialOrder σ' :=
  { syn := Syn m f
    toSyn : (σ' →_0 ℕ) ≈+ (Syn m f) := AddEquiv.refl (σ' →_0 ℕ)
    lo := LinearOrder.lift' (m.toSyn ∘ Finsupp.mapDomain f) ...
    ... }
```

It is easy to prove that remainder and Gröbner basis are preserved under a monomial order embedding. In general, a finite Gröbner basis not necessarily exists for an ideal in a polynomial ring with infinitely many variables, since such rings are not Noetherian. However, for a finitely generated ideal in $k[X_i | i \in \sigma]$, one can obtain a finite Gröbner basis by restricting to the subring generated by the variables appearing in a finite generating set, which is Noetherian. This basis can then be lifted to the original polynomial ring as a Gröbner basis in it.

Lemma 13. *Let $I \subseteq k[X_i | i \in \sigma]$ be a finitely generated ideal. Then I admits a finite Gröbner basis.*

```
lemma exists_isGroebnerBasis_finite_of_exists_span_finite
  {B : Set (MvPolynomial σ k)} (hB : B.Finite) :
  ∃ G : Set (MvPolynomial σ k), m.IsGroebnerBasis G (Ideal.span
    B) ∧ G.Finite
```

For an infinitely generated ideal $I \subseteq k[X_i | i \in \sigma]$, a Gröbner basis of I can be obtained by assembling Gröbner bases of its restrictions to finitely generated subrings. In [11, Proposition 1.8], this construction is performed along a prescribed ascending chain of subrings. We present below a modest generalization that is not limited to such an ascending chain.

Theorem 14. *Fix an ideal $I \subseteq k[X_i | i \in \sigma]$. Assume that for all finite $s \subseteq_{\text{fin}} \sigma$, there exist $\sigma_s \subseteq \sigma$ and $G_s \subseteq k[x_i]_{i \in \sigma_s}$ s.t. $s \subseteq \sigma_s$ and $G_s \subseteq k[x_i]_{i \in \sigma_s}$ is a Gröbner basis of $I \cap k[x_i]_{i \in \sigma_s}$. Then $\bigcup_{s \subseteq_{\text{fin}} \sigma} G_s$ is a Gröbner basis of I .*

```
theorem isGroebnerBasis_union_of_forall_finite_isGroebnerBasis_0
  {I : Ideal (MvPolynomial σ R)} {σ' : Finset σ → Type*}
  {m' : (s : Finset σ) → MonomialOrder (σ' s)}
  {e : (s : Finset σ) → (m' s).Embedding m}
  {G' : (s : Finset σ) → Set (MvPolynomial (σ' s) R)}
  (hG' : ∀ s, ∀ g ∈ G' s, IsUnit ((m' s).leadingCoeff g) ∨ g =
    0)
  (h : ∀ s, ↑s ⊆ Set.range (e s))
  (h' : ∀ s, (m' s).IsGroebnerBasis (G' s) (I.comap <| rename (e
    s))) :
  m.IsGroebnerBasis (⋃ s : Finset σ, rename (e s) '' G' s) I
```

The coverage condition means that every variable belongs to the stage subring eventually along F . Since every polynomial has finite variable support, each polynomial is eventually contained in these subrings. Neither the subrings nor their reduced bases need form an ascending chain.

Proof sketch. Let B be the reduced Gröbner basis of I . For $b \in B$, eventual coverage places b in the stage subring eventually. In every such stage, its monicity, minimal leading monomial, and standard tail identify it with an element of the stage's

reduced basis. Hence $B \subseteq \liminf_F G_j$. Conversely, let $h \in \liminf_F G_j$. Then $h \in I$, so some $b \in B$ satisfies $\text{LM}(b) \mid \text{LM}(h)$. Both b and h belong to G_j eventually. Nontriviality of F provides a stage where both memberships hold, and reducedness forces $h = b$. Therefore $\liminf_F G_j = B$. $\square$

In [11, Theorem 1.12], Iima and Yoshino characterize reduced Gröbner bases over polynomial rings with countably many variables via the limit inferior of reduced Gröbner bases on finite-variable subrings. We generalize this result to arbitrary index types using limit inferior constructions over filters.

Theorem 15. Fix an ideal $I \subseteq k[X_i \mid i \in \sigma]$, an index set ι , let $F \subseteq \mathcal{P}(\iota)$ be a filter over lattice $(\mathcal{P}(\iota), \subseteq)$ s.t. $\emptyset \notin F$. Assume for all $j \in \iota$, $\sigma_j \subseteq \sigma$ and G_j be the reduced Gröbner basis of ideal $I \cap k[x_i]_{i \in \sigma_j}$ on $k[x_i]_{i \in \sigma_j} \subseteq k[X_i \mid i \in \sigma]$. If $\bigcup_{s \in F} \bigcap_{j \in s} \sigma_j = \sigma$, then

$$\bigcup_{s \in F} \bigcap_{j \in s} G_j$$

is the reduced Gröbner basis of I .

```

theorem IsReduced.isReduced_liminf {k} [Field k] {I : Ideal (
  MvPolynomial σ k)}
{α : {σ' : α → Type*} {m' : (a : α) → MonomialOrder (σ' a)}
{f : Filter α} [inst_neBot : f.neBot]
{e : (a : α) → (m' a).Embedding m}
(hI : Set.univ = f.liminf (fun x ↦ Set.range (e x)))
{G' : (a : α) → Set (MvPolynomial (σ' a) k)}
(hG' : ∀ a, (m' a).IsGroebnerBasis (G' a) (I.comap <| rename (e
  a)))
(hG'' : ∀ a, (hG' a).IsReduced) :
  ∃ h : m.IsGroebnerBasis (f.liminf fun a ↦ rename (e a) '' G' a
    ) I, h.IsReduced

```

In particular, letting σ and ι be $\mathbb{N}$, F be the cofinite filter on $\mathbb{N}$, σ_n be $[0, n] \cap \mathbb{N}$, and G_n be the reduced Gröbner basis of $I \cap k[x_i]_{i \in [0, n] \cap \mathbb{N}}$, then

$$\bigcup_{n \in \mathbb{N}} \bigcap_{i \in A} G_i = \bigcup_{n \in \mathbb{N}} \bigcap_{i \geq n} G_i$$

is the reduced Gröbner basis of I [11, Theorem 1.12].

```

theorem IsReduced.isReduced_liminf_nat {k} [Field k]
{I : Ideal (MvPolynomial σ k)} {init : Nat} {e : ℕ ≈ σ}
{G' : (n : ℕ) → Set (MvPolynomial (Fin (n + init)) k)}
(hG' : ∀ n,
  (m.ofInjective <| e.injective.comp Fin.val_injective).
  IsGroebnerBasis
    (G' n) (I.comap <| rename (e o @Fin.val (n + init))))
(hG'' : ∀ n, (hG' n).IsReduced) :
  ∃ h : m.IsGroebnerBasis
    (Filter.cofinite.liminf fun i ↦ rename (e o @Fin.val (i +
      init)) '' G' i)
    I,
    h.IsReduced

```

5 Applications

5.1 Certified Polynomial Reasoning with Lean and CAS

Our formalization supports automated proofs of polynomial and ideal properties by combining computer algebra systems (CAS) with Lean. The tactics of Shen et al. [22] use

SageMath or SymPy to compute candidate Gröbner bases and polynomial certificates, then apply our library's verification theorems to reconstruct proofs. Every accepted conclusion is checked by Lean's kernel.

For concrete rational polynomials in finitely many variables, with lexicographic order, `gb_solve` certifies polynomial remainders, Gröbner-basis properties, and membership or non-membership in ideals and their radicals. The tactic `idealeq` proves ideal equality, while `add_gb_hyp` introduces a certified Gröbner basis for subsequent proofs. These tactics make the formal theory directly usable for a range of algebraic proof goals.

The library provides the common mathematical foundation for these tasks. Our remainder specification and Buchberger criterion certify polynomial reductions and Gröbner bases. Polynomial-combination identities establish membership, and certificates for both inclusions establish ideal equality. Non-membership uses a verified Gröbner basis and a nonzero reduced remainder, justified by our zero-remainder characterization. This approach supports certified symbolic computation without requiring an executable Gröbner-basis algorithm within the library: the CAS supplies computational witnesses, and the formalized theory establishes their consequences in Lean.

```

example :
  lex.IsRemainder (X 0 ^ 2 + X 1 ^ 3 + X 2 ^ 4 + X 3 ^ 5 :
    MvPolynomial (Fin 4) ℚ)
    {X 0, X 1, X 2, X 3} 0 := by
  simp only [← Set.range_get_singleton, ← Set.range_get_cons_list
    ]
  rw [IsRemainder.isRemainder_range_fintype, ← exists_and_right]
  use [X 0, X 1 ^ 2, X 2 ^ 3, X 3 ^ 4].get
  split_and
  focus
  set_option backward.isDefEq.respectTransparency false in
  simp [Fin.univ_succ, -List.get_eq_getElem, List.get]
  grind
  focus
  intro i
  fin_cases i
  all_goals {
    simp only [List.get, Fin.isValue]
    all_goals
      exact withBotDegree_le_of_repr_le <| by decide +kernel
  }
  simp

```

5.2 Formalizing Iima–Yoshino Problem 2.3

Guo et al. [9] use our library to formalize a solution to Iima–Yoshino Problem 2.3 over $\mathbb{C}$, together with a partition bijection underlying the first Rogers–Ramanujan identity. This application requires both infinitely many variables and an infinite Gröbner basis.

Work in $S = \mathbb{C}[x_1, x_2, \dots]$, with $\deg x_i = i$. Set $c = (\sqrt{5} - 1)/2$ and

$$(s_0, s_1, s_2, s_3, s_4) = (2, c, -1 - c, -1 - c, c),$$

reading subscripts modulo 5. For $n \geq 2$, define

$$g_n = (s_n - c)x_n + \frac{1}{2} \sum_{i=1}^{n-1} s_{n-2i} x_i x_{n-i},$$

and put

$$I = (g_n \mid n \geq 2, n \equiv 0, 2, 3 \pmod{5}),$$

$$G = \{g_{2m} \mid m \geq 1\} \cup \{c^{-1}g_{2m+1} \mid m \geq 1\}.$$

The application verifies that G is the reduced Gröbner basis of I for a suitable monomial order and establishes

$$S/I \cong \mathbb{C}[x_i \mid i \equiv \pm 1 \pmod{5}],$$

$$\langle \text{LT}(I) \rangle = \langle x_i^2, x_i x_{i+1} \mid i \geq 1 \rangle,$$

with a grading-preserving isomorphism. The initial ideal has infinitely many minimal generators, so no finite Gröbner basis can suffice for this order.

The two monomial bases encode partitions into parts congruent to 1 or 4 modulo 5 and partitions whose consecutive parts differ by at least two. In each weighted degree, canonical normal forms give an invertible matrix between these bases. The application formally extracts a perfect matching from its support, proving the partition form of the first Rogers–Ramanujan identity.

Our library supplies the Gröbner-basis and reducedness interfaces, the zero-remainder form of Buchberger’s criterion, and `existsUnique_isRemainder` for canonical normal forms. The application derives a standard-representation criterion from these results and verifies its hypotheses using five-periodic syzygies. Thus the same library supports both the infinite-basis proof and the normal-form construction used in the partition bijection.

The modules `CaseI/Statements.lean` and `ToGroebner/StandardRepresentation.lean` explicitly reference 14 distinct library declarations: 3 definitions and 11 lemmas or theorems (Table 1).¹ Each declaration is counted once; imports, comments, application-defined declarations, and Mathlib declarations are excluded. These counts measure direct interface reuse.

Table 1. Distinct library declarations explicitly referenced by the Iima–Yoshino application. Module names are relative to `Groebner/`.

Defining module	Definitions	Lemmas/theorems
<code>Remainder.lean</code>	1	2
<code>Groebner.lean</code>	1	5
<code>Reduced.lean</code>	1	1
<code>MonomialOrder.lean</code>	0	3
Total	3	11

¹Audited application revision `ffe3c0d7ef8d`, using library revision `69c476e02058`.

This application demonstrates reuse of the library for infinite-basis verification, reducedness, and canonical normal forms. It does not invoke the finite-subring embedding or filter-based limit interfaces.

6 Conclusion

We have formalized Gröbner basis theory in Lean 4 over fields with arbitrary variable types. Representing bases as sets and specifying remainders through finite witnesses allows the same definitions and proofs to accommodate finite and infinite bases. The development establishes Buchberger’s criterion and the existence and uniqueness of reduced bases. Monomial-order embeddings and filter-based reconstruction further connect the theory over different variable types and recover reduced bases from restrictions to subrings.

The two applications demonstrate how these interfaces support both computation and mathematical proofs. For finite-variable polynomials, external computer algebra systems supply certificates whose algebraic consequences are verified in Lean. For the Iima–Yoshino problem over $\mathbb{C}$, the library supports verification of an infinite reduced Gröbner basis and the normal-form argument yielding partition bijections for the first Rogers–Ramanujan identity. The infinite basis is essential in this example: its initial ideal has infinitely many minimal monomial generators. This gives a concrete use of the generality provided by arbitrary variable types and set-based Gröbner bases.

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